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21IPC501J
RESEARCH METHODOLOGY AND PUBLICATION ETHICS
Unit - 2

Syllabus :

- Preparation Of Project Proposal For Funding
- Identification Of Funding Agencies
- Project Proposals Structure
- Research Report Writing
- Data Visualization
- Communication Model – Audience Analysis;
- Identification Of Suitable Journal;
- Conference Presentation Types: Oral, Poster
- Difference In Audience Interaction
- Guideline For Effective Presentation

Syllabus :

- Synopsis And Thesis
- Extended Abstract, Graphical And Video Abstract
- Short Communications;
- Referencing Style And Tools For Referencing;
- Thesis Writing: Structure Preliminary Pages, Main Body, References;
- Evaluation Of Thesis – Examiner Reports
- Oral Defence.

SUBMISSION OF PROJECT PROPOSAL FOR FUNDING AGENACIES

Research Proposal:

- Research proposal is an overall plan scheme, structure and strategy designed to research questions.
- It should outline various tasks planned to fulfill research objectives.
- Reveals strengths and weaknesses of planned approach

1. Basic knowledge of funding agencies :

- Most funding agencies require research proposal in order to be considered for funding.
- Agencies often have annual deadlines and specific proposal guidelines
  - After deadline, proposals evaluated by agency staff and peer review committees
  - Evaluated and ranked on specific criteria
- If funded, researcher usually provides agency with annual or semiannual updates on research progress, and a final report. There are 3 questions will be asked by interviewer or research committee board

1. Is proposed problem significant? Is research appropriate to this funding agency?
2. Are the researchers qualified to conduct proposed research?
3. Are the budget, timelines, and resources proposed realistic to complete the research?

2. Organization of Proposal:

Page # 1:

Research Proposal – Title (60-120 Characters)

<Department Name>

<Full Postal Address of University>

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Organization of Proposal

Page # 2:

1. Project Title :________________________________
2. Principal Investigator :_________________________
3. Name, Designation:_____________________________
   Complete postal address:________________________
   Phone Number :_______________________________
   E-mail Id:____________________________________
4. Co-principal Investigator: (if applicable)
   Name, Designation:_____________________________
   Complete postal address:________________________
   Phone Number :_______________________________
   E-mail Id:____________________________________
5. Institution:_______________________
6. Project Summary :

(2 Paragraphs)

7. Keywords: (5 to 8)

Organization of Proposal

Page # 3 to 10 :

Technical details of the project may be expressed in analytical manner. It should be explained in detail as follows:

1. **Introduction – including a brief literature review**

- Introduction nothing but a shadow of the problem
- Origin of the proposal (specify year of the problem proposed)
- Definition of the problem
- Literature review – divided into 3 parts
  - Pioneering work, development work and recent development (till 2023)
- Status of research and developments in the subject (social objectives fulfillment)
  - International status
  - National status

Organization of Proposal

Page # 3 to 10 :

2. **Objectives / Research questions of the study**
   - Primary objectives
   - Subsidiary objectives – less than primary objectives
3. **Theoretical framework that underpins your study**
   - Main and associative objectives
4. **Concepted framework which constitutes basic of the study**
   - Complementary work to theoretical
5. **Hypothesis to be tested (if applicable)**
   - If you say 5 objectives that should be tested (tally with objective specific)
   - Statistical hypothesis – there is a significance difference in the relief time between two drugs curing an ailment.

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Page # 3 to 10 :

6. Research design adopted

- Characteristics of population, sample design and sample size
- Data collection instruments, pilot studies
- Data processing procedures (used for empirical studies)

7. Testing of the study

- Efficiency test, reliability test and experiment test
- Ethical issues involved in study and how to solve the issues (mostly in medical research)

8. Problems and limitations

9. Institutional resources

- Computers, library resources, other equipment

Organization of Proposal

Page # 3 to 10 :

10. Personnel

- Experts available with proposed investigation group/institution in the subject
- Description of people involved
- Demonstrate qualifications and distribution of work among personnel

11. Patent details

- Domestic and International patent details

12. Work plan

- Academic
- Administrative
- Methodology (design)
- Organization of work elements
- Time schedule of Suggested plan of action

Page # 3 to 10 :

13. Proposed time schedule for activities

- When all major project tasks will be conducted and completed
- Identifying all necessary tasks and creating realistic schedule for completion

1st Year	2nd Year	3rd Year	4th Year
Q1 …	Q1 …	Q1 …	Q1 …
Q2 …	Q2 …	Q2 …	Q2 …
Q3 …	Q3 …	Q3 …	Q3 …
Q4 …	Q4 …	Q4 …	Q4 …

14. Budge Estimate Summary

DIRECT COSTS* (note: no over projection	1st Year	2nd Year	3rd Year	4th Year
1. Personnel Salaries and Wages	.	.	.	.
2. Fringe benefits (recurring)	.	.	.	.
3. Consultants and Contracts	.	.	.	.
4. Travel	.	.	.	.
5. Supplies and materials	.	.	.	.
6. Communications (Telephone, postage, etc)	.	.	.	.
7. Equipment (Purchase)	.	.	.	.
8. Other (Equipment rental, etc.) overhead costs	.	.	.	.
TOTAL	.	.	.	.

Organization of Proposal

15. **Summary of the project (minimum 1 page)**
    - First and Second paragraph – primary objective
    - Third paragraph – subsidiary objective
16. **Anticipated results and conclusion**
    - Description of expected benefits
    - Re-emphasize context and value
17. **Annexure**
    - Any other relevant information
        - *Resumes of key personnel*
        - *Letters of support*
        - *Institutional descriptions*
    - Similar project details

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15. BIBLIOGRAPHY

- Harvard System – popular referencing system which lists books and periodicals
- DOI (Digital Object Identifier) – unique alphanumeric string assigned by a registration agency
- Journal article, book, conference paper, URL

Identification of Funding Agencies

- **1. Understand the Nature of Your Project**
- Before identifying a funding agency, clearly define the scope, goals, and outcomes of your project. This helps in targeting funders whose priorities align with your project.
- **Type of Project:** Research, social causes, startup development, education, community engagement, etc.
- **Field of Work:** Healthcare, technology, education, environmental science, etc.
- **Geographical Scope:** Local, national, international. Certain funders may focus on projects in specific regions.
- **Target Beneficiaries:** Understand who will benefit from your project, such as underprivileged communities, the general public, or scientific advancement.

Identification of Funding Agencies

- **2. Types of Funding Agencies**
- Funding agencies can vary significantly based on the type of project they support and their financial resources. Common types of funding agencies include:
- **Government Agencies:** Federal, state, and local governments offer a wide range of grants for public welfare, innovation, and research. These often include large funds for projects in public health, scientific research, and community development.
  - **Examples:**
    - **National Science Foundation (NSF)** – funds research in science and engineering.
    - **National Institutes of Health (NIH)** – supports biomedical research.
    - **European Research Council (ERC)** – funds innovative research across Europe.

Identification of Funding Agencies

- **Private Foundations:** Many philanthropic organizations provide grants to nonprofits or researchers. These tend to support specific causes, such as education, health, the arts, or social justice.
  - **Examples:**
    - **Bill & Melinda Gates Foundation** – focuses on global health, poverty alleviation, and education.
    - **Ford Foundation** – funds social justice, human rights, and economic opportunity initiatives.
- **Corporate Funding and Corporate Social Responsibility (CSR):** Many corporations support innovation and social initiatives, often through CSR programs or R&D grants. They may provide financial support for projects that align with their industry interests.
  - **Examples:**
    - **Google.org** – offers grants for tech-driven social impact projects.
    - **Pfizer** – funds healthcare and medical research initiatives.

Identification of Funding Agencies

- **Universities and Research Institutions:** Institutions may have internal grant programs, often for their own researchers or affiliated scholars. These can come from university endowments or partnerships with external agencies.
  - **Examples:**
    - **Harvard University Office of Sponsored Research** – manages research funding from a variety of sources.
    - **The Wellcome Trust** – focuses on biomedical research, often supporting university-led projects.
    - **Nonprofits and International Organizations:** Global organizations like the United Nations (UN) or World Health Organization (WHO) provide grants for development projects, often in developing countries or in sectors like health, education, and sustainability.

Identification of Funding Agencies

- **3. Where to Find Funding Agencies**
- There are multiple resources, both online and offline, that can help identify potential funding agencies:

- **Online Grant Databases:** These are searchable directories that provide detailed information on funding agencies, their grant programs, and application procedures. Popular platforms include:
  - Grants.gov – for U.S. federal grants.
  - Pivot – a research funding database for academic grants.
  - GrantForward – offers a comprehensive list of research grants and fellowships.
  - Candid (formerly Foundation Center) – offers directories of foundation grants.
- **Professional Networks:** Conferences, workshops, and academic associations often serve as platforms for learning about new funding opportunities. Researchers and project leaders can benefit by sharing insights and experiences with peers.

Identification of Funding Agencies

- **University Research Offices**: Many universities have offices that assist researchers in finding and applying for funding. These offices often subscribe to grant databases and maintain lists of funders.

- **Industry Associations and Trade Groups:** Depending on your sector, these organizations might provide information on corporate or industry-specific grants.

- **4. Match Project Goals with Funder Priorities**

- Each funding agency has its own mission and strategic focus. It is essential to align your project's goals with the priorities of potential funders. This requires understanding their:

- **Funding Priorities:** Many agencies focus on specific issues (e.g., cancer research, climate change) or certain types of projects (e.g., innovation, education, community welfare).

- **Eligibility Criteria:** Review who can apply – certain funders only provide grants to nonprofits, educational institutions, or researchers from specific regions or sectors.

Identification of Funding Agencies

- **Funding Amounts:** Funders vary widely in the size of grants they provide, from small seed grants to large, multi-year funding for major projects.
- **Grant Cycle:** Many funders have specific deadlines or cycles. Some accept applications year-round, while others have specific windows for submissions.

- **5. Stay Informed**

- New funding opportunities often arise. Regularly checking relevant websites, subscribing to newsletters, and staying connected within your professional community can keep you informed about emerging grants and opportunities.

- **1. Title of the report**
  - Title
  - Student name, e-mail id and supervisors/mentor’s name, email id
- **2. Abstract**
- A brief (150–300 words) summary of the entire research report.
- It includes:
  - **Purpose**: The main goal of the research.
  - **Methodology**: An overview of the research design and methods used.
  - **Results**: A snapshot of the key findings.
  - **Conclusion**: The overall implication of the findings.
- The abstract is often written last but appears first in the report.
- **3. Table of Contents**
- A list of all the major sections and subsections with page numbers.
- Helps the reader quickly navigate the report.

- **4. Introduction**
- The introduction sets the stage for the entire report by providing necessary background and defining the research scope. It typically includes:
- **Background**: Context of the research problem. Why is it important?
- **Problem Statement**: What specific issue is being addressed?
- **Objectives**: Clear goals or questions the research aims to answer.
- **Hypothesis** (if applicable): The assumption or expected outcome being tested.
- **Significance of the Study**: Why this research is valuable and relevant.
- **Scope and Limitations**: What the study will and won't cover, to clarify boundaries.

- **5. Literature Review**
- **Review of Existing Research**:
- Summarizes relevant prior studies to show the current state of knowledge on the topic.
- **Gaps in Literature**:
- Identifies where existing research is lacking and how your study fills these gaps.
- **Theoretical Framework**:
- The theories that guide or underpin the research.
- This section provides context and demonstrates familiarity with the field.

- **6. Methodology**
- A key part of any research report, this section describes in detail how the research was conducted:
- **Research Design**: Whether it's qualitative, quantitative, or mixed methods, and the rationale behind it.
- **Population and Sampling**: Who or what is being studied, and how the sample was chosen.
- **Data Collection Methods**: Instruments, tools, and procedures used (surveys, interviews, experiments, etc.).
- **Data Analysis Techniques**: Statistical or qualitative methods used to analyze the collected data.
- **Ethical Considerations**: Any ethical issues addressed, such as consent and confidentiality.
- The methodology should be thorough enough to allow other researchers to replicate the study.

- **7. Results**
- **Presentation of Data**: Raw data or summarized data, presented in an objective, neutral manner.
  - **Tables, Charts, and Graphs**: Visual aids to help interpret complex data. These should be labeled and referenced in the text.
- **Findings**: Main findings based on the data. No interpretation or discussion of the meaning of the results yet; just the facts.
- **Unexpected Results** (if any): Any surprising findings that were not anticipated during the research.

- **8. Discussion**
- **Interpretation of Results**: Analyzes the results in the context of the research objectives and questions.
- **Comparison with Previous Research**: How do the findings align with or differ from prior studies?
- **Implications**: The significance of the findings for the field, policy, or practice.
- **Limitations of the Study**: What factors may have affected the results, and how might these be addressed in future research?
- **Future Research Suggestions**: Directions for further investigation.

- **9. Conclusion**
- A succinct summary of the key findings and their implications.
- Restate the research objectives and briefly explain how the research has addressed them.
- Emphasize the main contributions of the study and why they matter.
- **10. Recommendations (optional)**
- Actionable suggestions based on the research findings. This section is important in applied or policy-driven research.
- May include practical steps for implementing the research or exploring additional areas.

- **11. References**
- A complete list of all sources cited in the research report.
- Follow a specific citation style (APA, MLA, Chicago, etc.), ensuring consistency.
- Only include sources that are directly referenced in the report.

- **12. Appendices**
- Additional material that supports the report but is too bulky for the main text.
- This might include:
  - Raw data tables
  - Questionnaires or surveys
  - Detailed calculations
  - Supplemental figures or documents
- Each appendix should be labeled (Appendix A, Appendix B, etc.) and referenced in the main body.

Purpose of Data Visualization

- **Simplifying Complex Data**: Large datasets can be overwhelming. Visualization makes it easier to comprehend by summarizing key insights.
- **Identifying Patterns and Trends**: Visual representations can reveal patterns, trends, or outliers that may not be easily noticeable in textual or numerical data.
- **Communication**: Well-designed visualizations make it easier to communicate findings to a wide audience, including those who may not be familiar with the data.
- **Comparing Variables**: Visual tools like bar charts, scatter plots, or line graphs allow for clear comparison between variables.
- **Facilitating Decision-Making**: Visuals help stakeholders understand data quickly and make informed decisions based on the insights.

Data Visualization

- Common Types of Data Visualization in Research
  - Bar Charts
  - Histograms
  - Line Charts
  - Pie Charts
  - Scatter Plots
  - Geographical Maps

Tools for Data Visualization

- **Spreadsheet Software-**Tools like Microsoft Excel or Google Sheets
- **Statistical Software-**Tools like SPSS, SAS, and R
- **Data Visualization Software-**Platforms like Tableau, Power BI, or Google Data Studio
- **Programming Languages-**R (with ggplot2 package), Python (with Matplotlib and Seaborn libraries)
- **Visualization Libraries-**JavaScript libraries like D3.js and Plotly

Best Practices in Data Visualization

- Clarity and Simplicity
- Choose Appropriate Visualizations
- Consistent Scales
- Use Color Effectively
- Focus on Key Insights
- Interpretation Aid

Challenges in Data Visualization

- **Misleading Visualizations**: Poorly designed visualizations can distort the truth. For instance, manipulating the axis scales can exaggerate or minimize differences.
- **Data Overload**: Including too much data in one visualization can confuse rather than clarify.
- **Bias**: It’s important to present data objectively without bias. Choosing specific visualization types or manipulating visual scales to support a particular narrative can mislead viewers.

Basic Components of the Communication Model

- **Sender (Researcher):** The researcher, who has conducted the study, is responsible for delivering the message.
- **Message (Research Content):** The content or findings of the research which need to be communicated.
- **Encoding:** The process of structuring and formatting the message in a way that the audience can comprehend.

Channel (Medium of Communication):

- The medium through which the message is communicated. In research, such as written, oral, visual, digital

Communication Model and Audience Analysis

- **Receiver (Audience):**
  - The audience for whom the research is intended. This could include scholars/academicians, practitioners, general Public.
- **Decoding:**
  - The audience's interpretation of the message.
- **Feedback:**
  - Feedback allows the researcher to evaluate the effectiveness of the communication and make adjustments if needed.
- **Noise (Barriers to Communication):**
  - Technical Jargon:
  - Cultural or Contextual Differences
  - Physical Barriers

Audience Analysis in Research Communication

- **Audience analysis** is the process of identifying and understanding the characteristics, expectations, and needs of the audience for whom the research communication is intended.

- **Steps in Audience Analysis**

  a) Identifying the Audience

  - Experts
  - Non-experts
  - Stakeholders
  - Peers and Colleagues

Audience Analysis in Research Communication

- **Understanding Audience Knowledge Level**
  - High-level audience
  - General audience
- **Anticipating Audience Expectations**
  - Novelty
  - Clarity
  - Credibility
- **Addressing Potential Questions or Concerns**

Identification of Suitable Journal

- Identifying the **suitable journal** for your research is a crucial step in the publication process. Publishing in the right journal enhances the visibility, impact, and credibility of your work.

- **1. Understand Your Research**
  - **a. Define Your Research Area and Scope**
    - **Subject Matter:**
    - **Disciplinary Boundaries.**
    - **Research Type:**
- **b. Determine the Novelty and Significance**
  - **Originality**
  - **Impact**

Identification of Suitable Journal

- **Identify Potential Journals**
- **a. Explore Journals in Your Field**
  - **Literature Review:** Examine the journals where key papers in your area are published.
  - **References:** Look at the reference lists of relevant articles to identify frequently cited journals.
- **b. Use Journal Finder Tools**
  - **Elsevier Journal Finder:** Elsevier Journal Finder
  - **Springer Journal Suggester:** Springer Journal Suggester
  - **IEEE Publication Recommender:** IEEE Publication Recommender
  - **Edanz Journal Selector:** Edanz Journal Selector
- These tools allow you to input your manuscript's title, abstract, or keywords to receive journal suggestions based on content similarity.

Identification of Suitable Journal

- **3. Evaluate Journal Scope and Aims**
- **a. Review the Journal’s Aims and Scope**
- **Aims Statement:** Understand the journal’s objectives and the type of research it seeks to publish.
- **Scope:**
- Ensure your research aligns with the journal’s thematic areas and subject focus.
- **b. Analyse Published Articles**
- **Content Alignment:**
- Check recent issues to see if similar studies have been published.
- **Quality and Style:**
- Assess the quality, writing style, and formatting of articles to match your work accordingly.

Identification of Suitable Journal

- **4. Assess Journal Reputation and Impact**
- **a. Impact Factor and Metrics**
- **Impact Factor:** A measure reflecting the yearly average number of citations to recent articles published in the journal. Higher impact factors often indicate greater prestige.
- **Other Metrics:** Consider metrics like **h-index**, **SCImago Journal Rank (SJR)**, and **Eigenfactor**.
- *Resources:*
- **Journal Citation Reports (JCR):** Clarivate JCR
- **Scimago Journal & Country Rank:** ScimagoJR

Identification of Suitable Journal

- **b. Publisher Reputation**

- **Established Publishers:** Journals published by reputable publishers (e.g., Elsevier, Springer, Wiley, Taylor & Francis, IEEE) are generally more trustworthy.

- **c. Indexing and Abstracting**

- **Databases:** Ensure the journal is indexed in major databases like **PubMed**, **Scopus**, **Web of Science**, or **IEEE Xplore**, which enhances discoverability.

Identification of Suitable Journal

- **Review Submission Guidelines and Policies**
- **a. Manuscript Requirements**
  - **Formatting:** Adhere to the journal’s formatting guidelines regarding structure, citation style, word count, etc.
  - **Sections:** Ensure your paper includes all required sections (e.g., abstract, keywords, references).
- **b. Open Access vs. Subscription-Based**
  - **Open Access (OA):** Articles are freely accessible to all. Some journals offer OA options for a fee.
  - **Subscription-Based:** Access is restricted to subscribers. Authors may not have the option to make their articles freely available.

Identification of Suitable Journal

- Considerations:
  - **Funding Requirements:** Some funding bodies require research to be published OA.
  - **Visibility:** Articles generally receive higher visibility and citation rates.
- **c. Publication Fees and Charges**
  - **Article Processing Charges (APCs):** Fees required for publishing, especially in OA journals.
  - **Waivers:** Check if the journal offers fee waivers for authors from low-income countries or with financial constraints.

Identification of Suitable Journal

- **d. Peer Review Process**
  - **Type of Review:** Understand whether the journal uses single-blind, double-blind, or open peer review.
  - **Review Timeline:** Consider the average time from submission to decision, especially if timely publication is important.
- **e. Acceptance Rate**
  - **Competitiveness:** High acceptance rates may indicate lower selectivity, while low rates suggest higher competition and prestige.

Identification of Suitable Journal

- **Final Checklist Before Submission**

1. **Relevance:** Confirm that your research aligns with the journal's scope and audience.
2. **Formatting:** Ensure your manuscript adheres to all formatting and submission guidelines.
3. **Quality:** Review the quality of your writing, data presentation, and overall manuscript.
4. **Ethics:** Verify that your research complies with ethical standards and that all necessary approvals are obtained.
5. **Permissions:** Secure permissions for any third-party content included in your manuscript.
6. **Cover Letter:** Prepare a compelling cover letter tailored (suitable) to the journal, highlighting the significance of your work.

Identification of Suitable Journal

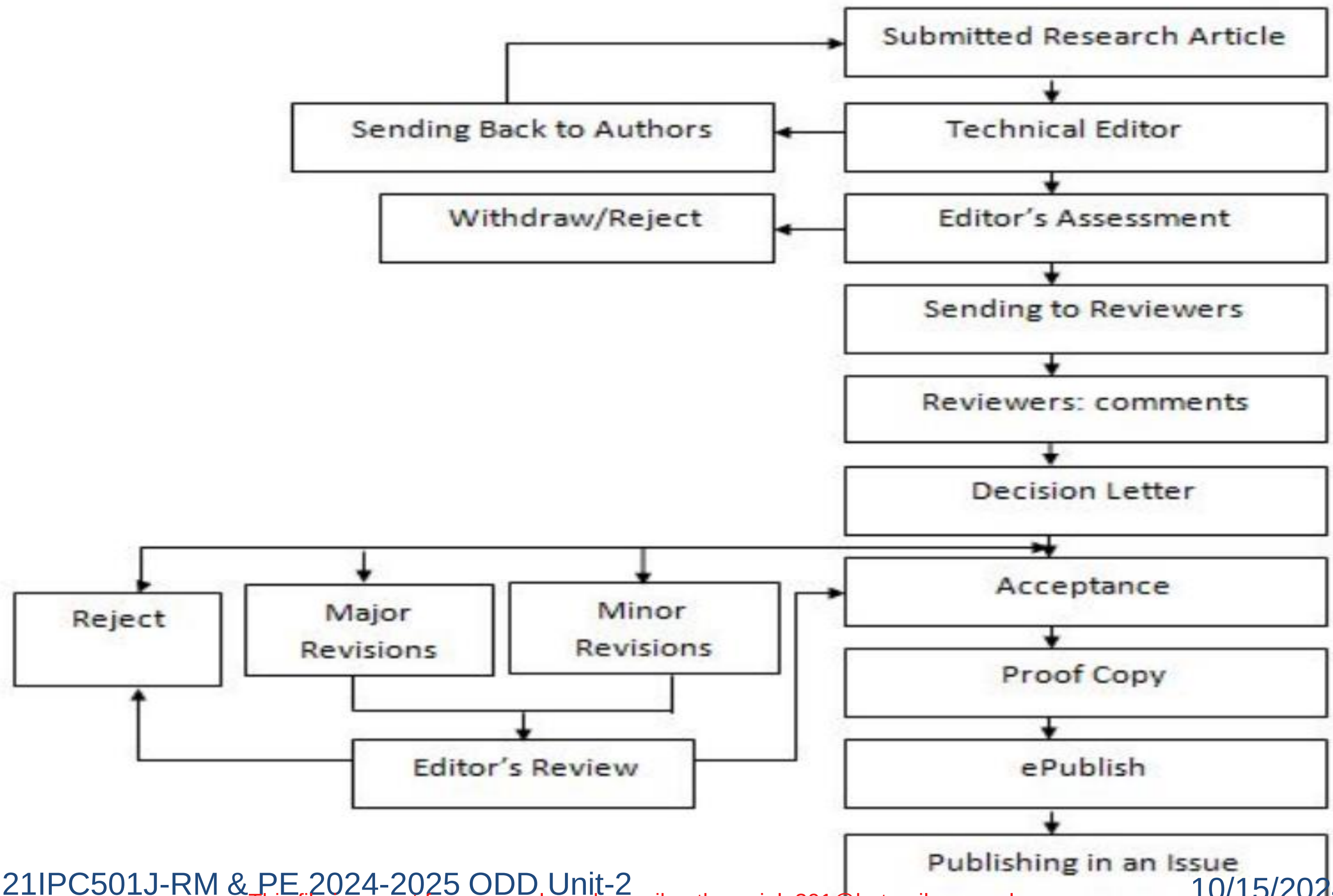
- **Tips for Successful Journal Selection**
- **a. Start Early:** Begin the journal selection process during the early stages of your research to align your study with potential publication venues.
- **b. Be Realistic:** Match your manuscript's quality and novelty with the journal's standards and impact level.
- **c. Diversify Options:** Prepare a shortlist of target journals ranked by preference, including fall-back options in case of rejection.
- **d. Understand the Journal's Audience:** Tailor your writing and presentation to resonate with the journal's readership.
- **e. Follow Ethical Guidelines:** Ensure compliance with all ethical standards and avoid any potential conflicts of interest.

Identification of Suitable Journal

- **Common Mistakes to Avoid**
- **a. Ignoring Journal Scope**
  - Submitting to journals whose scope does not align with your research can lead to immediate rejection.
- **b. Overlooking Impact and Reach**
  - Choosing journals with limited reach may reduce the visibility and impact of your research.
- **c. Disregarding Submission Guidelines**
  - Non-compliance with formatting and submission rules can delay the review process or result in outright rejection.
- **d. Chasing High Impact Journals Prematurely**
  - Attempting to publish in top-tier journals without sufficient novelty or rigor can lead to repeated rejections. Build your publication record progressively.

- **<u>After Identifying Potential Journals</u>**
- **a. Prepare Your Manuscript Accordingly**
  - Align your manuscript's structure, language, and presentation with the chosen journal's guidelines.
- **b. Write a Compelling Cover Letter**
  - Highlight the significance, originality, and relevance of your research to the journal's aims.
- **c. Seek Feedback Before Submission**
  - Have peers or mentors review your manuscript to ensure it meets the journal's standards and is free from errors.

Roadmap after submission



DIFFERENT ASPECTS TO BE FOLLOWED IN SUBMISSION OF RESEARCH ARTICLES FOR PUBLICATION

- Familiarize yourself with potential publications
  - Read academic journals related to your field of study
  - Search online for published research papers, conference papers, and journal articles
  - Ask a colleague or professor for a suggested reading list
- Choose the publication that best suits your research article
- Prepare your research article
- Ask a colleague and professor to review your research article
- Revise your article
- Submit your article
- Keep trying.

ORAL PRESENTATION

- **Visual Aids**
  - Maps, Photos, film clips, graphs, diagrams, and charts can enhance a presentation.
- **Handouts**
  - Handouts consists of name, title, data, abstract, outline and bibliography of references .

ORAL PRESENTATION

- **Delivery Tips**
  - Adequate eye contact- size of the audience, never let them out of your sight
  - Interpersonal communication
  - Professionalism and enthusiasm
  - At the end, summarize main points and have a confident with clear conclusion.
- **Fear and Nervousness**
  - Accept nervousness for what it is
  - A good preparation will increase your self-confidence
- **Evaluation**
  - Noting down important points arising out of discussion
  - Keep a possible evaluation in mind

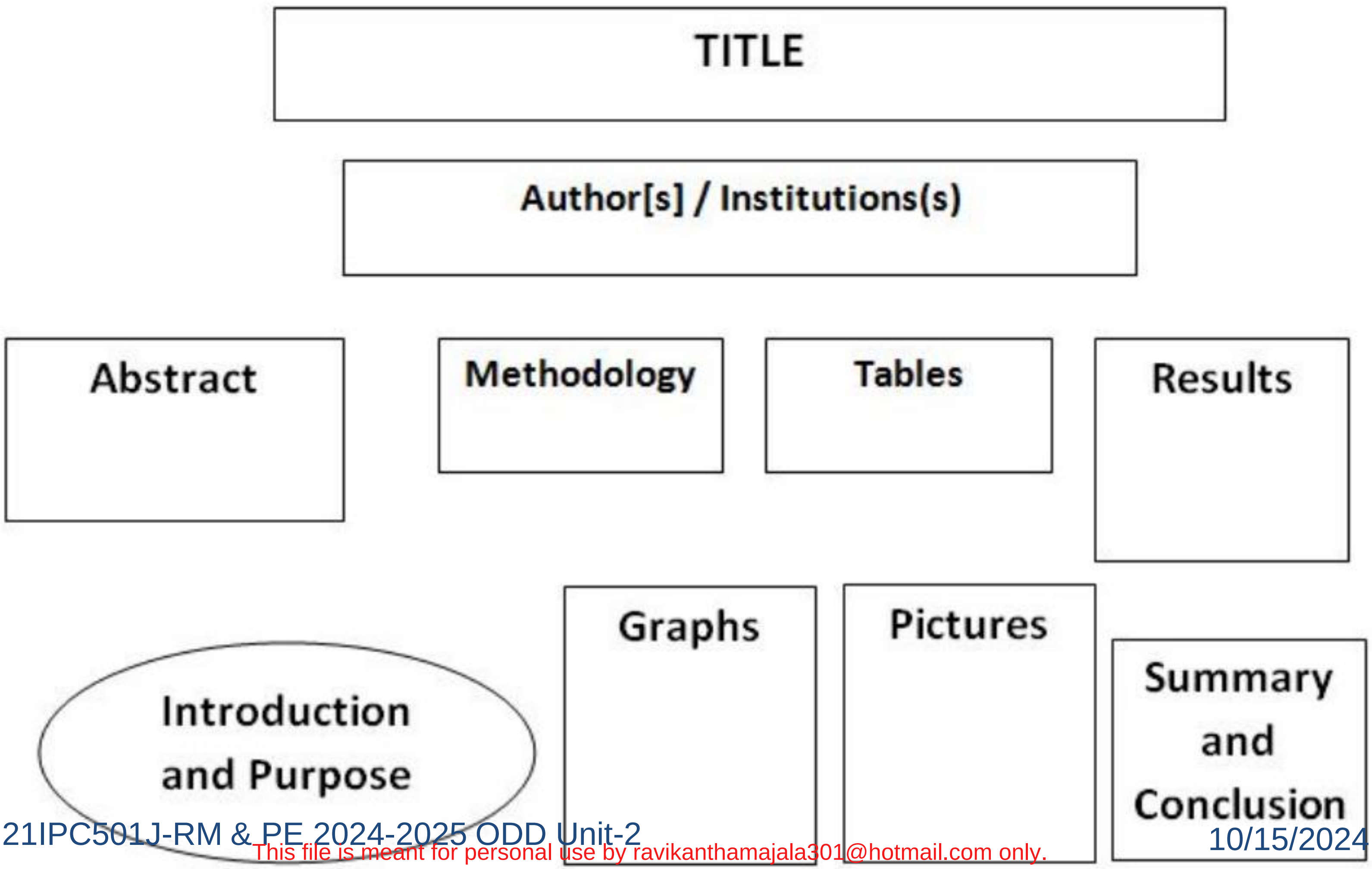
POSTER PRESENTATION

- Poster presentation advertises your project.
- It combines text and graphics to present research information in an interesting and accessibly way
- A poster presentation allows you to interact one-on-one with people interested in your research
- Poster sessions allow for 2 hours or more of discussion with interested visitors

Characteristics of good posters:

- Posters are permanent, portable and well-sequenced (Ordered)
- Posters can be simple or very elaborate (Focused)
- Posters can be used alone or in a series to tell a story (Graphic)

POSTER PRESENTATION LAYOUT



POSTER CONTENT

- Paper size : A0 (1180 x 841 mm)
- Title (96 pt)
- Authors(s) / Institution(s) (72 pt)
- Abstract (36 pt)
- Introduction and Purpose (36 pt)
- Methodology (24 pt)
- Tables (24 pt)
- Graphs (24 pt)
- Pictures (24 pt)
- Data and Results (24 pt)
- Summary and Conclusion (24 pt)
- References and Acknowledgements (optional) (18 pt)

POSTER PRESENTATION GUIDELINES:

- Keep text brief
- Keep figures simple (a picture is work a thousand words)
- Should be readable to someone standing at a distance of 6 feet
- Use appropriate blank space between words, sections and figures
- Horizontal posters will only fit the boards (4-ft. high x 8-ft. wide free standing)

LIMITATIONS OF POSTER PRESENTATION

- Posters tend to contain too much detail
- Transporting them can be difficult
- The more elaborate posters require extensive preparation and can be quite costly

TAILORING PRESENTATION TO TARGET AUDIENCE

- **Effective presentation attributes:**
- Content – relevant, Informative, Illustrative, Factual and Observation
- Presenters Voice Delivery – Tone, Modulate, Pause, volume and Language
- Presenter – Animated, Body language, Attire, Listening and Seeing

SIX STEP APPROACH FOR SUCCESSFUL PRESENTATION

1. Plan – Market analysis and fundamental thinking
2. Organize – framework, skeleton of the package
3. Support – material developed
4. Stage – presenter identifies facilities, equipment, and schedules
5. Deliver – Show time, convey information, ideas and propositions
6. Follow-up – take care of loose ends, apply lessons learned toward a better job

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Do’s and Don’ts in presentations:

Sl. No.	Do’s	Don’ts
1	Leave no doubt the presentation is finished	Apologize
2	Perhaps tie back to attention getting line at the start: answer a question	Trail off
3	Return to opening joke or story	Introduce new info or ideas
4	Consider a proverb or quotation	Read it, make it too long
5	Maintain eye contract	Change the mood or style

Synopsis

- **Purpose of a Synopsis:**
  - **Proposal and Approval**: The synopsis is usually required when applying for research approval from an academic or research committee, such as a university's review board or ethics committee. It allows reviewers to assess the research's feasibility, importance, and alignment with academic standards.
  - **Research Plan**: It acts as a blueprint for the research, offering a roadmap of the researcher's objectives, methodology, and expected outcomes.
  - **Clarification of Research Focus**: The synopsis ensures that the research question is clear, the methodology is sound, and the scope is well-defined.

Structure of a Synopsis:

1. **Title**:
   - A clear, concise (short), and descriptive title that reflects the research problem or question.
2. **Introduction/Background**:
   - Briefly explains the context of the research topic, providing background information.
   - Highlights the significance of the study and why the research problem is worth investigating.
3. **Research Problem**:
   - States the specific issue or gap in knowledge that the research seeks to address.

Synopsis

4. **Research Objectives/Questions**:
   - Clearly outlines the objectives or research questions that guide the study.
   - These should be specific, measurable, and aligned with the research problem.
5. **Literature Review (Brief)**:
   - A summary of key literature related to the research topic.
   - Identifies the gaps in the existing literature that the research will address.
6. **Hypothesis (if applicable)**:
   - In studies where hypotheses are tested, a synopsis includes the proposed hypothesis or hypotheses that the research seeks to confirm or reject.

Synopsis

7. **Methodology**:
   - Details the proposed research design, data collection methods, and tools.
   - Includes the sample size, sampling techniques, instruments, procedures, and analysis techniques.

8. **Scope and Limitations**:
   7. Explains the boundaries of the study—what the research will cover and what it will exclude.
   - Identifies potential limitations and challenges that may arise during the research process.

9. **Expected Outcomes**:
   - Provides a brief description of what the researcher hopes to achieve through the study.
   - Can include theoretical contributions, practical applications, or advancements in a specific field.

10. **References**:

- A list of key academic references that have informed the research idea and methodology.

- **Length of a Synopsis:**

- A synopsis is usually between **2,000 to 3,000 words** (depending on institutional guidelines), offering a concise but comprehensive overview of the proposed research.

Thesis

- A **thesis** is a comprehensive, written document submitted as part of a degree requirement, such as a Master's or PhD program.

- **Purpose of a Thesis:**

- **Demonstrate Research Skills**: A thesis showcases the researcher's ability to design and conduct independent, rigorous research, interpret results, and draw meaningful conclusions.

- **Contribute to Knowledge**: It makes an original contribution to the existing body of knowledge in a specific field of study.

- **Degree Requirement**: The submission and defence of a thesis is typically a requirement for completing a Master's or Doctoral program.

Thesis

- **Structure of a Thesis:**

1. **Title Page**:
   - Contains the title of the thesis, the author's name, institutional affiliation, degree for which it is submitted, supervisor's name, and submission date.
2. **Abstract**:
   - A succinct summary (usually 200-300 words) of the thesis, including the research problem, methodology, key findings, and conclusions.
   - Gives readers a quick overview of the study's purpose and results.
3. **Acknowledgments**:
   - A section where the researcher acknowledges the contributions of those who helped or supported the research, including advisors, funding agencies, and participants.

Thesis

4. **Table of Contents**:
   - An organized list of all chapters, sections, and sub-sections with corresponding page numbers.
5. **Introduction**:
   - Provides a detailed introduction to the research topic, including the problem statement, research objectives, research questions, and significance of the study.
   - Establishes the context of the study, sets out its scope, and explains its potential contributions to the field.
6. **Literature Review**:
   - A critical analysis and synthesis of existing literature on the topic.
   - Demonstrates the researcher's awareness of previous studies, highlights gaps in the literature, and situates the current research within the broader scholarly conversation.

Thesis

7. **Research Methodology**:

- Describes in detail how the research was conducted, including:
  - **Research Design**: Qualitative, quantitative, or mixed-methods design.
  - **Sampling**: Sample size, selection criteria, and sampling techniques.
  - **Data Collection**: Methods used for collecting data (e.g., surveys, interviews, experiments, archival research).
  - **Data Analysis**: Statistical or qualitative techniques used to analyze the data.
  - **Validity and Reliability**: Discusses the accuracy and trustworthiness of the research instruments and data.
  - **Ethical Considerations**: Explains the ethical framework under which the research was conducted.

8. **Results/Findings**:

- Presents the key findings of the research, organized systematically.
- Data may be presented using charts, graphs, tables, or other visual aids.
- For quantitative studies, statistical results are often emphasized; for qualitative research, themes and patterns are discussed.

9. **Discussion**:

- Interprets the results in relation to the research questions or hypotheses.
- Explains the significance of the findings, their implications for theory and practice, and how they relate to the existing literature.
- Identifies any limitations in the research and suggests areas for future research.

10. **Conclusion**:

- Summarizes the main findings and arguments of the thesis.
- Highlights the contributions of the research to the field.
- May include recommendations for policy, practice, or future research.

11. **References/Bibliography**:

- A complete list of all sources cited in the thesis, following a specific citation style (e.g., APA, MLA, Chicago).

12. **Appendices**:

- Contains supplementary material, such as raw data, questionnaires, consent forms, or additional charts, which were not included in the main body but are relevant to the research.

Thesis

Aspect	Synopsis	Thesis
Purpose	Proposal or plan of the research.	Comprehensive documentation of the completed research.
Timing	Submitted before the research begins or early in the process.	Submitted after the research is completed.
Length	2,000 to 3,000 words (brief overview).	50,000 to 100,000 words (in-depth analysis).
Content	Summary of research problem, objectives, methodology, and expected outcomes.	Full description of the research problem, methodology, results, and discussion.
Focus	Primarily on planning and justification.	Primarily on presenting the research findings and contribution.
Review	Evaluated to determine whether the research can proceed.	Evaluated for the quality of research and contribution to the field.
Outcome	Approval of the research proposal.	Award of a degree upon successful defense of the thesis.

Tips for writing a Strong Thesis

- **Start Early**:
  - Writing a thesis is a long process. Begin drafting chapters (such as the introduction and literature review) early in the research process to avoid last-minute rushes.
- **Follow Institutional Guidelines**:
  - Every institution has its specific guidelines for thesis formatting, length, citation styles, and submission deadlines. Make sure to follow these carefully.
- **Maintain Clear and Logical Flow**:
  - The thesis should be well-structured, with each section logically flowing into the next. Readers should easily follow your research process from the introduction to the conclusion.

Tips for writing a Strong Thesis

- **Use High-Quality Visuals**:
  - For results sections, present data in easily interpretable formats, such as tables, graphs, and charts. Visual aids enhance understanding and make the thesis more engaging.
- **Revise and Edit**:
  - The first draft is never the final one. Thoroughly revise your thesis multiple times, focusing on clarity, coherence, and academic rigor.
- **Seek Regular Feedback**:
  - Continuously share your progress with your supervisor and incorporate their feedback. Regular communication helps avoid major revisions at the end.

Extended Abstract

An **extended abstract** is a longer version of a traditional abstract, providing more detailed information about the research.

❑ Purpose of an Extended Abstract:

- **Comprehensive Overview**: Unlike traditional abstracts (typically 200-300 words), extended abstracts provide more detailed insight into the research, including more comprehensive information about the methods, results, and conclusions.

- **Proposal for Conference Presentations**: Researchers often submit extended abstracts for conference presentations (oral or poster). These abstracts are reviewed to decide whether the research merits a presentation.

- **Brief Substitute for Full Paper**: In some cases, extended abstracts serve as a stand-in for a full paper, especially in conference proceedings where publishing the entire paper may not be feasible.

Graphical Abstract

- **Purpose of a Graphical Abstract:**
- **Visual Summary**:
- Graphical abstracts condense the research into a single image or visual graphic, providing an immediate snapshot of the work's essence.
- **Increased Engagement**:
- They are designed to be visually engaging, making the research more accessible to wider audiences, including those outside the academic community.
- **Enhances Online Visibility**:
- In digital platforms and scientific databases, graphical abstracts can enhance the visibility and dissemination of research by attracting readers who might not otherwise engage with the text-heavy content.

Video Abstract

❑ Purpose of a Video Abstract:

- **Engage a Broader Audience**:
- Video abstracts are designed to be engaging and accessible, allowing the researcher to communicate their work to a broader audience, including those outside the academic community.
- **Enhance Understanding**:
- By using audio-visual elements, video abstracts can explain complex concepts in a clearer and more engaging way than text alone.
- **Increase Online Visibility**:
- Video abstracts can be shared on platforms like YouTube, academic journals, social media, and personal websites, increasing the visibility of the research.

Comparison of Abstract

Aspect	Extended Abstract	Graphical Abstract	Video Abstract
Purpose	Provide a detailed summary of the research	Visual summary for quick understanding	Engage and explain research through video
Format	Text-based with sections like a short paper	Visual (image or diagram)	Video (2-5 minutes) with narration or animations
Length	1,000-2,000 words	Single image or visual	2-5 minutes
Audience	Academics, reviewers	Academics, general public	Broad audience (academics and non-experts)
Accessibility	Moderate (requires reading)	High (visual-based, quick to grasp)	High (engaging and easy to share)
Key Strength	Detailed description of the research	Instant comprehension through visuals	High engagement, multimedia communication

Referencing Style

Common Referencing Styles

1. **APA (American Psychological Association) Style**:
   - **Field**: Commonly used in the social sciences, psychology, education, and nursing.
   - **Format**: Author-date format.
   - **In-text Citation**: (Author, Year). For example: (Smith, 2020).
   - **Reference List**: Full citations are listed alphabetically by the author's last name.
   - **Example**:
     - Book: Smith, J. (2020). *Title of the book*. Publisher.
     - Journal Article: Smith, J. (2020). Title of the article. *Title of the Journal*, *Volume*(Issue), page range. DOI or URL.

Referencing Style

2. **MLA (Modern Language Association) Style**:
   - **Field**: Commonly used in humanities, particularly in literature, philosophy, and the arts.
   - **Format**: Author-page format.
   - **In-text Citation**: (Author Page). For example: (Smith 23).
   - **Works Cited**: List of references at the end of the document, arranged alphabetically by the author's last name.
   - **Example**:
     - Book: Smith, John. *Title of the Book*. Publisher, Year.
     - Journal Article: Smith, John. "Title of the Article." *Title of the Journal*, vol. 10, no. 2, Year, pp. 23-45.

Referencing Style

3. **Chicago/Turabian Style**:

- **Field**: Widely used in history, arts, and some social sciences.
- **Format**: Two systems - Notes and Bibliography (often used in humanities) and Author-Date (used in sciences).
- **In-text Citation (Notes)**: Superscript number linked to footnotes or endnotes.
- **Bibliography**: Full citations listed at the end of the document.
- **Example**:
  - Book (Notes and Bibliography): Smith, John. *Title of the Book*. City: Publisher, Year.
  - Journal Article: Smith, John. "Title of the Article." *Title of the Journal* 10, no. 2 (Year): 23-45.

Referencing Style

4. **IEEE (Institute of Electrical and Electronics Engineers) Style**:
   - **Field**: Commonly used in engineering and technical fields.
   - **Format**: Numeric citation style.
   - **In-text Citation**: [1], [2], etc. For example: [1].
   - **Reference List**: References are numbered in the order they are cited in the text.
   - **Example**:
     - Book: [1] J. Smith, *Title of the Book*, 2nd ed. City: Publisher, Year.
     - Journal Article: [2] J. Smith, "Title of the Article," *Title of the Journal*, vol. 10, no. 2, pp. 23-45, Year.

Referencing Style

5. **Harvard Style**:

- **Field**: Widely used in various disciplines, particularly in the UK and Australia.
- **Format**: Author-date format.
- **In-text Citation**: (Author Year). For example: (Smith 2020).
- **Reference List**: Entries are listed alphabetically by author.
- **Example**:
  - Book: Smith, J. 2020. *Title of the Book*. Publisher.
  - Journal Article: Smith, J. 2020. Title of the article. *Title of the Journal*, *Volume*(Issue), page range.

Thesis Writing Style

1. Preliminary Pages

- The preliminary pages are critical in establishing the context and framework for the thesis. Each element serves a specific purpose.

a. Title Page

  - Title of the thesis: Should be concise yet descriptive.
  - Author’s name: Your full name.
  - Institution: Name of your university or college.
  - Department: Your specific department or program.
  - Submission date: Month and year of submission.

- **Purpose**: Provides essential identification information and establishes the official nature of the document.

Thesis Writing Style

- **b. Abstract**
  - Purpose of the study.
  - Research questions or hypotheses.
  - Methodology: Brief description of how the research was conducted.
  - Key findings: Summarize the main results.
  - Conclusions: State the significance and implications of the findings.
- **Purpose**: To offer a quick overview of the research, helping readers decide whether to read the full thesis. It should be succinct, typically 250-500 words.
- **Acknowledgments**
  - Gratitude to advisors, faculty members, funding organizations, friends, and family.
  - Specific contributions (e.g., intellectual guidance, emotional support).

Thesis Writing Style

- **Table of Contents**
  - Main sections and chapters, with corresponding page numbers.
  - Subsections and appendices, if applicable.
- **Purpose**: Helps readers quickly locate information within the thesis, enhancing navigability.
- **List of Figures and Tables**
  - Titles of figures and tables, along with page numbers.
- **Purpose**: Provides a quick reference for visual elements included in the thesis, making it easier for readers to find specific data representations.

Thesis Writing Style

- **2. Main Body**
- The main body is the heart of the thesis and is structured to guide the reader through your research findings in a logical manner.
- **a. Introduction**
  - Context: Briefly outline the background of the research topic.
  - Research Problem: Clearly define the problem being addressed.
  - Research Objectives: What you aim to achieve with the research.
  - Research Questions/Hypotheses: Specify the questions your study seeks to answer.
  - Significance of the Study: Explain why the research is important and what contribution it makes to the field.
- **Purpose**: Establishes the framework for the entire thesis and prepares the reader for what to expect.

Thesis Writing Style

- **b. Literature Review**
  - Overview of key literature: Summarize existing research related to your topic.
  - Themes and Gaps: Identify major themes, methodologies, and gaps in the literature that your research addresses.
  - Theoretical Framework: Discuss relevant theories that underpin your research.
- **Purpose**: Demonstrates familiarity with existing research and provides context for your study, establishing the rationale for your research.

- **c. Methodology**
  - Research Design: Outline the overall approach (e.g., qualitative, quantitative, mixed-methods).
  - Participants/Subjects: Describe the sample size, selection criteria, and demographics.
  - Data Collection Methods: Explain how data was collected (e.g., surveys, interviews, experiments).
  - Data Analysis: Describe the techniques used to analyze the data (e.g., statistical analysis, thematic analysis).
  - Ethical Considerations: Discuss how ethical issues were addressed (e.g., informed consent, confidentiality).
- **Purpose**: Provides enough detail for replication and establishes the validity of the research design.
- Demonstrates familiarity with existing research and provides context for your study, establishing the rationale for your research

- **d. Results**
  - Presentation of Findings: Clearly present the data collected (e.g., tables, charts, graphs).
  - Descriptive Statistics: Include summaries that highlight key characteristics of the data.
  - Inferential Statistics: Present analyses that answer the research questions, including p-values and confidence intervals if applicable.
- **Purpose**: Objectively presents the findings without interpretation, allowing readers to draw their own conclusions based on the data.

Thesis Writing Style

- **e. Discussion**
  - Interpretation of Results: Discuss the implications of the findings in relation to the research questions and existing literature.
  - Comparison with Previous Research: How do your results align or contrast with earlier studies?
  - Limitations: Acknowledge any weaknesses or limitations in your study (e.g., sample size, methodology).
  - Future Research: Suggest areas for further investigation based on your findings.
- **Purpose**: To analyse and interpret the results, demonstrating their significance and contextualizing them within the broader field of research.

Thesis Writing Style

- **f. Conclusion**
  - Summary of Key Findings: Briefly recap the major discoveries of the research.
  - Contributions to Knowledge: Discuss how your research advances understanding in the field.
  - Final Thoughts: Conclude with reflections on the research process and its impact.
- **Purpose**: To provide a clear and concise ending that emphasizes the importance of the research and leaves a lasting impression on the reader.

Thesis Writing Style

3. References

- The references section is crucial for academic integrity and scholarly communication.
- **a. Reference List/Bibliography**
  - Full citations for all sources cited in the thesis.
  - Format according to the required citation style (e.g., APA, MLA, Chicago).
  - Organize entries alphabetically by the author's last name or numerically if following a specific citation style.
- **Purpose**: Allows readers to locate the original sources used in the research, demonstrating scholarly rigor and respect for intellectual property.

Thesis Writing Style

4. Appendices (if applicable)

- Appendices contain supplementary material that is relevant but not integral to the main text.
- **a. Components of Appendices:**
  - Raw data: Detailed datasets that support the findings.
  - Questionnaires: Copies of surveys or interview questions used.
  - Supplementary Analyses: Additional analyses that support the findings but are too lengthy for the main body.
  - Visual Materials: Any additional figures or illustrations not included in the main body.

Purpose: Provides readers with additional context or information that enhances understanding without interrupting the flow of the main thesis.

Evaluation of Thesis

1. Structure of Examiner Reports

- While the exact format may vary depending on institutional guidelines, most examiner reports typically include the following sections:
- **a. Introduction**
  - **Purpose**: A brief overview of the thesis topic and research questions.
  - **Content**: Summary of the research aims, significance, and context.
- **b. Evaluation Criteria**
  - **Purpose**: Outline the criteria used for evaluating the thesis.
  - **Content**: May include originality, research design, methodology, data analysis, literature review, clarity, and presentation.

Evaluation of Thesis

- **c. Detailed Comments**

- **Purpose**: Provide comprehensive feedback on specific aspects of the thesis.
- **Content**: This section usually includes:
  - **Originality and Contribution**: Assessment of the uniqueness of the research and its contributions to the field.
  - **Methodology**: Evaluation of the appropriateness and rigor of the research design and methods used.
  - **Results**: Commentary on the clarity and validity of the findings presented.
  - **Discussion**: Analysis of how well the findings are interpreted and contextualized within existing literature.
  - **Writing Quality**: Feedback on the organization, clarity, and academic writing style.

Evaluation of Thesis

- **d. Summary of Strengths and Weaknesses**
- **Purpose**: Highlight key strengths and weaknesses of the thesis.
- **Content**: A concise overview that encapsulates the main points made in the detailed comments.
- **e. Recommendations**
- **Purpose**: Suggest actions based on the evaluation.
- **Content**: Recommendations may include:
  - Approval of the thesis with minor or major revisions.
  - Suggestions for further research.
  - Comments on areas needing significant improvement.

Purpose of Examiner Report

- Examiner reports serve several important functions in the thesis evaluation process:
- **a. Quality Assurance**
- Ensures that the thesis meets established academic standards and contributes meaningfully to the field.
- **b. Constructive Feedback**
- Provides the candidate with valuable insights that can help improve their work, whether through revisions or in future research endeavours.
- **c. Decision-Making**
- Assists the evaluation committee in making informed decisions about the outcome of the thesis defense (e.g., pass, conditional pass with revisions, or fail).
- **d. Documentation**
- Creates an official record of the evaluation process, outlining the rationale behind the examiners' decisions and comments.

Oral Defense

- **1. Purpose of the Oral Defense**
- The oral defense serves several key purposes:
- **a. Validation of Research**
  - **Assessment**: It provides an opportunity for examiners to evaluate the originality, quality, and significance of the research.
  - **Clarification**: Candidates clarify their methodology, results, and conclusions in response to questions.
- **b. Evaluation of Understanding**
  - **Knowledge Check**: Examines the candidate's depth of understanding of the research topic and related literature.
  - **Critical Thinking**: Tests the candidate's ability to think critically and articulate their thoughts under pressure.

Oral Defense

- **c. Feedback Opportunity**
  - **Constructive Criticism**: Offers a platform for receiving feedback from experts, which can be beneficial for refining the thesis or planning future research.
- **d. Development of Communication Skills**
  - **Presentation Skills**: Enhances the candidate's ability to present complex information clearly and concisely.
  - **Q&A Handling**: Improves the candidate's skills in responding to challenging questions and defending their work.

Evaluation of Oral Defense

- Examiners typically evaluate the oral defense based on several criteria:
- **a. Clarity and Organization:** Assess the clarity of the presentation and the logical flow of information.
- **b. Depth of Knowledge:** Evaluate the candidate's understanding of their research topic, methods, and related literature.
- **c. Ability to Defend Research:** Gauge how well the candidate responds to questions and defends their methodology, findings, and conclusions.
- **d. Presentation Skills:** Observe the candidate's communication skills, including their ability to engage the audience and convey complex ideas effectively.
- **e. Professionalism:** Consider the candidate's demeanor, confidence, and professionalism throughout the defense.

Outcomes of the Oral Defense

- The outcome of the oral defense can vary:
- **a. Pass**
- **Approval**: The thesis is accepted as is, or with minor revisions suggested.
- **b. Conditional Pass**
- **Revisions Required**: The thesis is approved pending specific revisions or corrections as outlined by the examiners.
- **c. Fail**
- **Re-evaluation Needed**: The candidate may need to address significant deficiencies in the thesis or retake the defense after further preparation.

SRM INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

Schematic Chart for a Researcher when he undergoes Ph.D. in an academic Institute

1. Ph.D. Application
2. Ph.D. - Written Test
3. Ph.D. - Interview
4. Ph.D. - Selection
5. Ph.D. - Joining Process
6. Ph.D. - DC Meeting
7. Course Work

1. Extensive Reading
2. Discussion with Supervisor
3. Conference / Workshop
4. Discussion with experts
5. DC Meeting Provisional Registration
6. Publishing research articles
7. Pre PhD

1. Synopsis Meeting
2. Synopsis Submission
3. Ph.D. Thesis Submission
4. National , International Expert Report
5. Ph.D. VIVA
6. Ph.D. Degree

Thank you